

Cmpm125 Graphics Assignment 3

Team:

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Team Project: Particle System + Shader Graph

Theme: Magic Effects

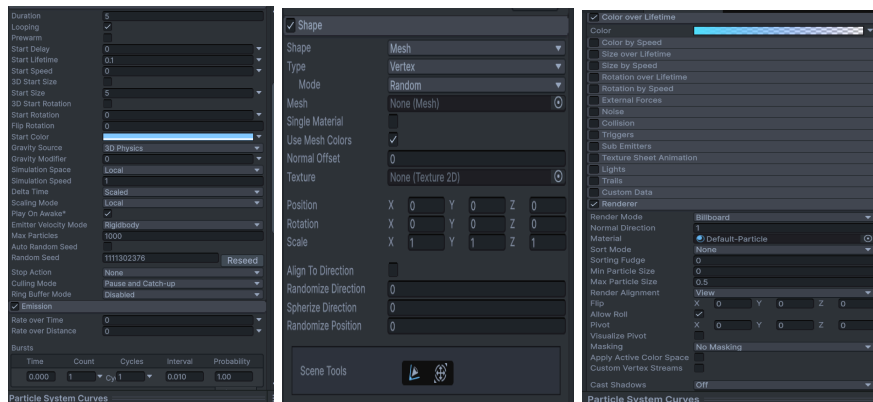
Project Notes:

1. To interact in the demo scene, use the A and D keys to move left and right.
2. Our magic effects incorporate 2 Particle Systems and 1 Shader Graph.

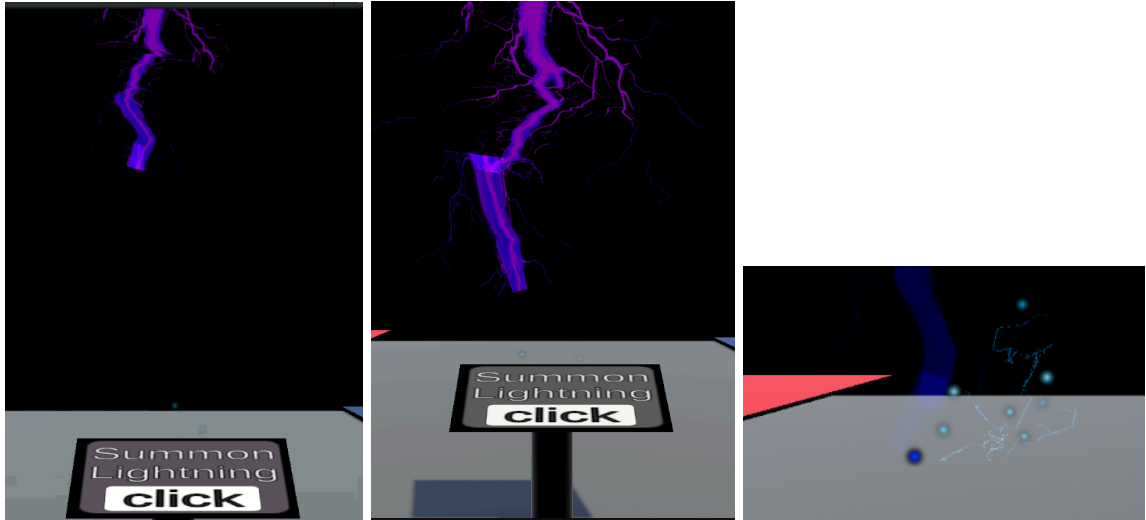
1: Magic Particle Explosion (Particle System) A blue and purple particle effect that shoots our particles in all directions



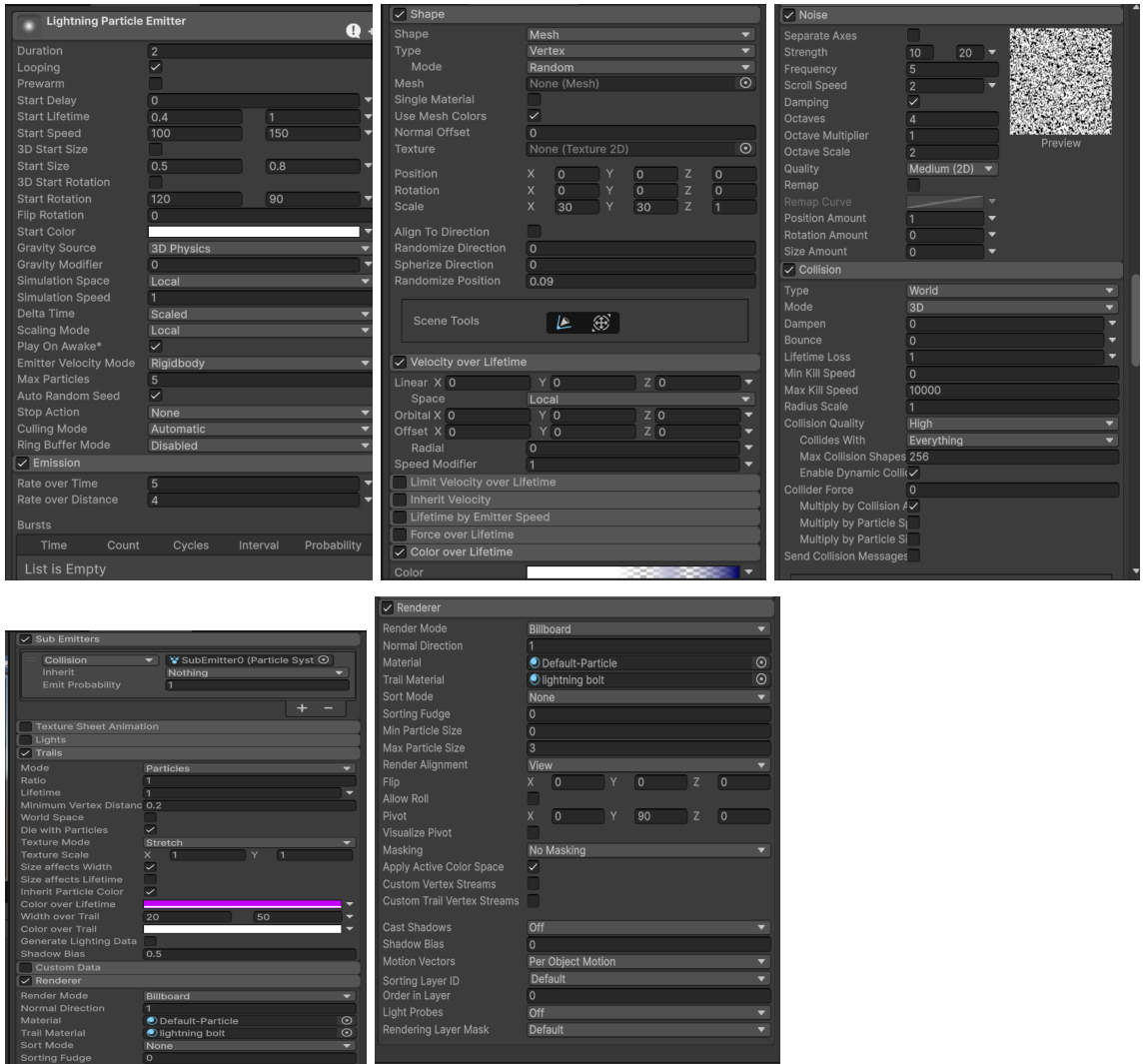
Particle Explosion Particle System Configuration :



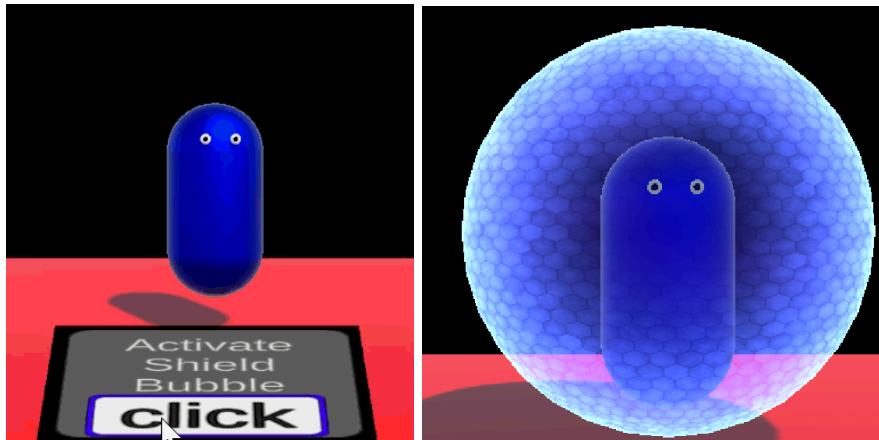
2: Magic Lightning Bolt (Particle System): A blue and purple lightning effect that shoots multiple lightning bolts towards a target. This Particle system includes sub-emitters and collision, as shown by the small blue sparks that generate when the lightning strikes a surface.



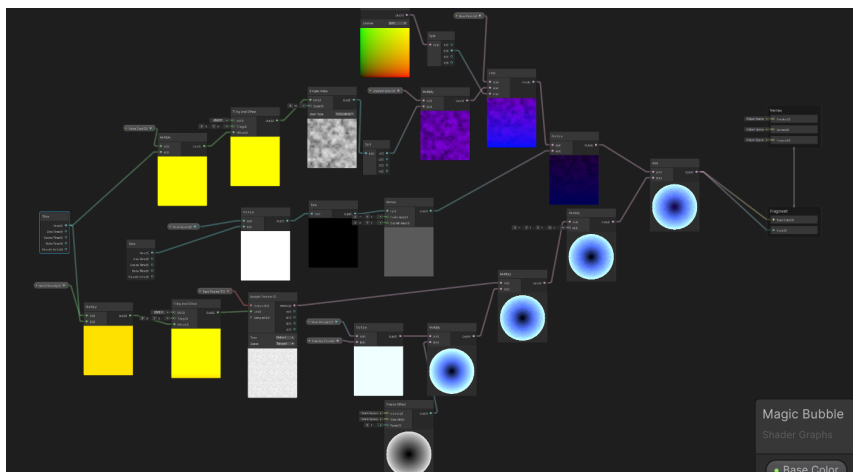
Lightning Particle System Configuration:



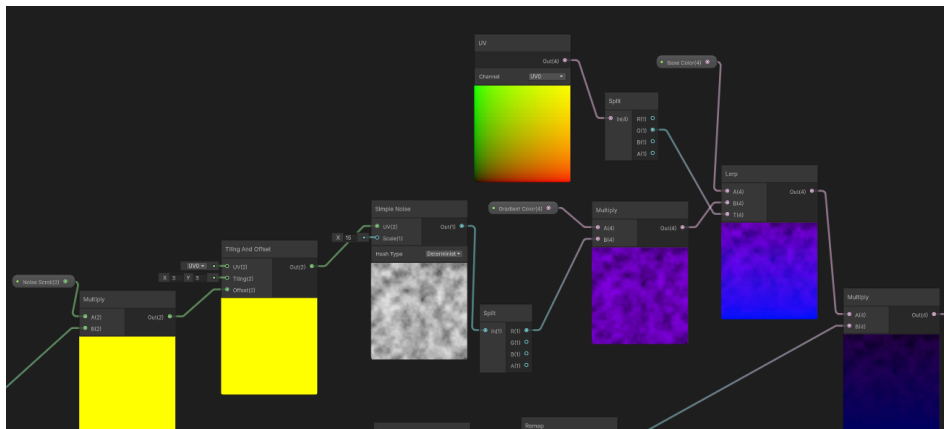
3: Magic Shield Bubble: (Shader Graph) A blue and purple hexagon magic shield bubble created through a shader graph.



Annotated Shader Code:



Segment 1: Scrolling Noise Gradient



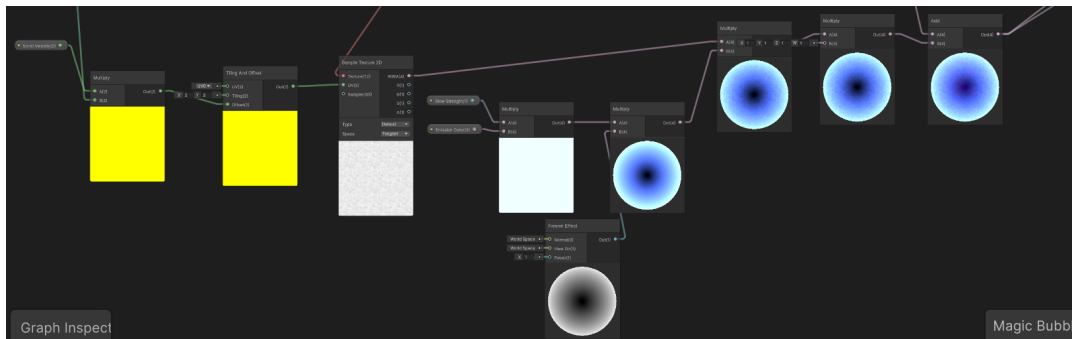
This section of the shader graph introduces an offset-tiling node that shifts as the underlying perlin noise module. I then multiplied the shifting perlin noise by a purple-and-blue gradient mapped to the texture's UV coordinates. This creates a scrolling gradient noise texture that is applied as the base layer of the Shield Bubble.

Section 2: Pulsating Glow Effect



In this section of the Shader Graph, I created a pulsing effect that, when combined with the gradient noise, pulses the underlying shader to go from being on and off. This effect uses a time node combined with a sine wave node to create the pulsing effect.

Section 3: Fresnel + scrolling hexagon texture:



The final part of the shader graph is the outer hexagon shield effect. In this section, there are two main parts. The first part on the left takes a hexagon grid texture and scrolls it along the x-axis. The second part of this section is the Fresnel effect, which produces the blue glow surrounding the shield bubble. When combined, these two sections make up the outermost layer of the shield bubble.

Contributions:

Michael: For this project, I was responsible for creating both the magic shield bubble shader graph and the lightning particle system. I also implemented the interactable scene, which allows the user to control and interact with the various particle/shader graph systems. For this project, I learned a lot about working with shader code. Even though I followed some basic instructional videos on how to implement some of the different visual effects, I still found that making each effect unique and combining them all together still required a lot of trial and error. While I do

find shader graphs to be significantly easier than shader code, it still requires a lot of playing around with different settings and values to produce something that looks visually appealing.

Vivian: Vivian created the particle explosion particle system.